POSTER

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ZHIXING HUANG, Victoria University of Wellington, Wellington, WGN, New Zealand

BING XUE, Victoria University of Wellington, Wellington, WGN, New Zealand

MENGJIE ZHANG, Victoria University of Wellington, Wellington, WGN, New Zealand

JEREMY S ROONEY, University of Otago, Dunedin, OTA, New Zealand

KEITH C GORDON, Callaghan Innovation, Lower Hutt, New Zealand

DANIEL P KILLEEN, Plant & Food Research, New Zealand, Auckland, AUK, New Zealand

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Zhixing Huang✉, Bing Xue, Mengjie Zhang
{zhixing.huang,bing.xue,mengjie.zhang}@ecs.vuw.ac.nz
the Centre for Data Science and Artificial Intelligence &
School of Engineering and Computer Science
Victoria University of Wellington, New Zealand

Jeremy S. Rooney
jeremy.rooney@otago.ac.nz
Department of Chemistry, University of Otago
New Zealand

Keith C. Gordon
keith.gordon@otago.ac.nz
MacDiarmid Institute for Advanced Materials and Nanotechnology, Chemistry Department, University of Otago, New Zealand

Daniel P. Killeen
daniel.killeen@plantandfood.co.nz
The New Zealand Institute for Plant and Food Research Limited, New Zealand

Abstract

Analyzing spectral data is an effective and efficient way to estimate the nutrition of food products. However, spectral data are usually high-dimensional and noisy, which makes spectral data analysis challenging. Data augmentation is an effective method to enhance machine learning models in analyzing spectral data. However, most existing data augmentation methods are designed by human experts, which is tedious and requires extensive expertise. To improve the effectiveness of data augmentation and reduce the dependency on domain knowledge, this paper proposes a genetic programming method to design data augmentation methods automatically. The proposed genetic programming method mimics the real distribution and produces new data instances by adding offsets to the original data. We take a fish spectroscopic dataset as an example to verify the effectiveness of the proposed method. The empirical results show that the data augmentation method designed by genetic programming has a very competitive performance compared to the state-of-the-art manually designed ones and provides good interpretability.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; *Machine learning*.

Keywords

Data Augmentation, Genetic Programming, Spectroscopic Data

ACM Reference Format:

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1 Introduction

Spectral data analysis is an important task in chemical industries and food production. However, the spectral data is usually high-dimensional as it covers a wide range of spectrum. The spectral data usually comes with a lot of noise because of instrument bias, glass vials, and many other spectral data artefacts (e.g., light scatter effects). In addition, we commonly have a limited number of data in spectral data analysis as it is expensive and time-consuming to label the ground truth of spectral data by chemical anlaysis methods. To effectively analyze our on-hand spectral data, we need data augmentation (DA) techniques.

DA is an effective pre-processing technique to enhance the robustness of machine learning models. One of the DA methods in spectral data analysis is to synthesize new data instances by manipulating original data instances with random noises. This is usually implemented by a function, and we call this function as data augmentation function (DAF) in this paper. A DAF accepts an original spectral data instance and returns a new one. The DAF usually manipulates random values sampled from certain distributions, such as uniform and Gaussian distributions. The random values enable DAFs to synthesize different new data instances in each execution. However, existing DAFs are mostly designed by human experts and relatively simple. They might be specialized for certain domains and targets, which limits the effectiveness of the existing DAFs.

This paper proposes to design effective DAFs by genetic programming (GP) [7]. GP is a popular symbolic search technique that searches for effective symbol combinations within a given solution space. Using GP to design DAFs has two main advantages over other methods. First, the symbolic solutions of GP allow human users to understand, analyze, and adapt DAFs intuitively, facilitating further studies on the DAFs. Second, existing studies of GP have shown that GP is less dependent on a large number of training data than many existing machine learning methods, which is highly fit with DA tasks. In this paper, we expect GP to help us find more effective or at least competitive DAFs than the manually designed ones. Specifically, we adopt linear genetic programming (LGP) [2], a variant of GP methods, for evolving DAFs here. It has been shown that LGP can evolve more effective and compact programs than canonical tree-based GP [5] and can naturally produce multiple outputs [4, 6].

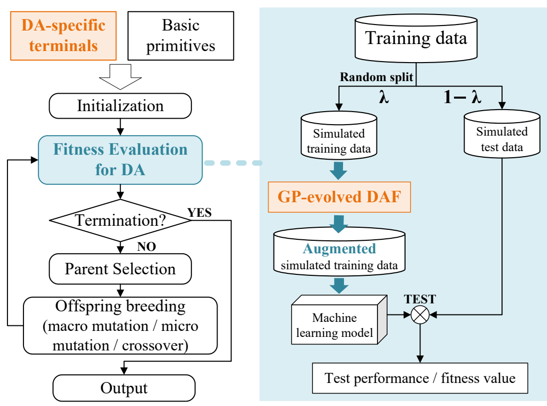

Figure 1: LGP Evolutionary framework (left) and the proposed fitness evaluation for DA (right).

This work aims to predict fish biomass compositions based on spectral data, which is a regression task in machine learning. Specifically, we design a set of DA-specific terminals for GP to design new DAFs and propose a fitness evaluation process to evaluate the effectiveness of the new DAFs. The major contributions of this paper are two-fold:

(1) This work shows that automatically designing DA methods by GP is better than or at least competitive with designing by human experts in improving machine learning performance. It can be extended to emerging domains and new datasets where we have insufficient expertise for designing DA methods.
(2) This work shows that GP methods are capable of designing generative models. The generative models accept the original data and random values as inputs and produce arbitrary number of unseen data. More importantly, the generative models designed by GP are interpretable to humans, allowing further analyses and applications.

2 Proposed Method

2.1 Evolutionary Framework

This paper proposes to evolve DAFs for fish spectral data by LGP. The evolutionary framework of the proposed method follows the one of basic LGP, as shown in the left part of Fig. 1. We first initialize a population of LGP individuals based on the given primitives, each individual representing a DAF. The given primitives include the newly proposed DA-specific terminals and the basic ones. The LGP population evolves a certain number of generations and produces new LGP individuals in each generation. The new LGP individuals are produced from the selected parents by a set of genetic operators (e.g., macro mutation and crossover). The individuals with better fitness are more likely to be selected as parents. The genetic operators produce new individuals by 1) inserting or removing random instructions, 2) varying LGP instructions, and 3) swapping instruction segments between two parents. All the individuals are evaluated via the newly proposed fitness evaluation process of DA. Finally, the best-of-the-run individual is outputted for unseen data.

2.2 Proposed Fitness Evaluation for DA

This section develops a new fitness evaluation process for DA, denoted as DAFE. The main idea of DAFE is to simulate the application of DAFs. The right part of Fig. 1 shows an illustrative schematic of DAFE. We first randomly split the given training set into two subsets, one called the simulative training set and the other called the simulative test set. The simulative training and test sets account for λ and $1-\lambda$ of the training set, respectively. The designed DAF augments the simulated training set. A machine learning model is trained on the augmented simulative training set and further tested on the simulative test set. The test performance of the machine learning model is treated as the fitness values of the designed DAF.

The designed DAF synthesizes new data instances by adding offsets to the original data. Denote an designed DAF as $\rho(\cdot):\mathbb{R}\mapsto\mathbb{R}$ and a spectral data instance as $\mathbf{x}\in\mathbb{R}^m$ where m is the dimension of the spectral data, then we have the augmented data instance $\hat{\mathbf{x}}$:

$$\hat{\mathbf{x}} = \mathbf{x} + \Delta(\mathbf{x}),$$

$$\Delta(\mathbf{x}) = [\rho(\mathbf{x}_1), \rho(\mathbf{x}_2), ..., \rho(\mathbf{x}_m)]^T.$$

$\rho(\mathbf{x}_i), i\in[1..m]$ produce the offset for each element of $\mathbf{x}$. $\rho(\mathbf{x}_i)$ outputs a range of different values for the same input $\mathbf{x}_i$ by including inherent random numbers in its execution. The inherent random number samples a random value from a uniform distribution.

To ensure the range of the offset is reasonable, we scale the $\rho(\mathbf{x}_i)$ to be $\widehat{\rho(\mathbf{x}_i)}$ before adding to $\mathbf{x}$. α is predefined based on the maximum noise range of spectral data. Denote $\Delta\mathbf{x}_i = \rho(\mathbf{x}_i) - \mathbf{x}_i$, we have

$$\widehat{\rho(\mathbf{x}_i)} = (\rho'(\mathbf{x}_i) * \alpha + \texttt{rand})/2,$$

$$\rho'(\mathbf{x}_i) = \frac{\mathbb{I}(\Delta\mathbf{x}_i)(|\Delta\mathbf{x}_i| - \min_{i\in[1..m]}|\Delta\mathbf{x}_i|)}{\max_{i\in[1..m]}|\Delta\mathbf{x}_i| - \min_{i\in[1..m]}|\Delta\mathbf{x}_i|},$$

where $\mathbb{I}(\Delta\mathbf{x}_i)$ returns 1 if $\Delta\mathbf{x}_i > 0$, returns -1 if $\Delta\mathbf{x}_i < 0$, and returns 0 otherwise.

In this paper, we repeat DAFE 5 times to have an average performance for each designed DAF. For each DAFE, we randomly split 20% training data as simulated training data and the rest as simulated test data ($\lambda = 0.2$) to encourage the designed DAFs to guess the unknown distributions. The maximum offset α is set as 0.05.

2.3 DA-specific Terminals

We propose a set of DA-specific terminals for designing DAFs for spectral data. The DA-specific terminals allow DAFs to produce offsets based on the spectral data and their distribution. The DA-specific terminals include:

(1) `rand`: It is the inherent random number in $\rho(\mathbf{x}_i)$. It returns a random number ranging from $[-\alpha, \alpha]$.
(2) $\mathbf{x}_i$: It returns $\mathbf{x}_i$ itself.
(3) `cursor`: It is the position of $\mathbf{x}_i$ over the wavenumber range of the data instance. `cursor`$=\frac{i}{m} - 0.5$.
(4) `avg5`$(\mathbf{x}_i)$: It is the average value over $\mathbf{x}_i$ to $\mathbf{x}_{i+4}$ (or $\mathbf{x}_i$ to $\mathbf{x}_m$ if $i+4>m$. The follows are the same).
(5) `std5`$(\mathbf{x}_i)$: It is the standard deviation over $\mathbf{x}_i$ to $\mathbf{x}_{i+4}$.
(6) `max5`$(\mathbf{x}_i)$: It is the maximum value over $\mathbf{x}_i$ to $\mathbf{x}_{i+4}$.
(7) `min5`$(\mathbf{x}_i)$: It is the minimum value over $\mathbf{x}_i$ to $\mathbf{x}_{i+4}$.

(8) avg(cursor): It is the average value over all the $\mathbf{x}_i$ in the training data.
(9) std(cursor): It is the standard deviation over all the $\mathbf{x}_i$ in the training data.

3 Experiment Design

3.1 Fish Spectroscopic Dataset

This paper takes a fish spectroscopic dataset as a case study to verify the proposed method. The fish spectroscopic dataset consists of 117 spectroscopic instances[1] from 39 unique fish samples, covering two kinds of deep-water fish spices in New Zealand, Hoki and Mackerel. Each spectroscopic instance has 427 input features. The dataset records the biomass compositions (e.g., water and protein) for each instances. We predict the biomass compositions of fish samples based on the spectroscopic data, which is a regression task in machine learning. To comprehensively evaluate the performance of the proposed method over the dataset, we perform six-fold cross-validation for 10 times with unique random seeds.

3.2 Comparison Design

In this experiment, we apply the proposed GP method to evolve DAFs for six biomass compositions (i.e., water, ash, nitrogen, protein, fat, and lipids yield) in the fish spectroscopic dataset, respectively. For each independent run, LGP trains DAFs based on five folds of data and outputs the best DAF for testing on the one fold of unseen data. The output DAF augments the training data (the five folds of data) for five times, and a linear regression model is trained on the augmented training data. Subsequently, the trained linear model predicts a certain biomass target on the unseen test data. The predicting performance (i.e., regression performance) of the machine learning model is represented by R^2.

We compare the performance of the designed DAFs with another DAF specifically designed for spectral data by human experts [1], denoted by SDAF. SDAF is a popular DA method for spectral data [13]. The settings of SDAF in our experiments follow the recommended settings in [1]. We apply SDAF to augment the training data and then train and test the same machine learning model to obtain the test performance of SDAF.

4 Experiment Results

4.1 Training and Test Performances

This section investigates the training and test performance of the DAFs designed by LGP in each run. We denote the training and test performance of LGP as "LGPtrain" and "LGPDA", respectively, and compare them with the test performance of SDAF. There are two major observations. First, the DAFs designed by LGP have a competitive performance with the manually designed DAF. In five of the six biomass, the DAFs by LGP have a higher median of R^2 than SDAF, implying that LGP successfully discovers more effective DAFs than human experts in most cases. Second, we perform the Wilcoxon rank-sum test with an α of 0.05 to compare the test performance of LGP and SDAF. The p-values show that LGP has a significantly better R^2 than SDAF on fat biomass with a p-value of 0.013.

[1]InGaAs Raman at 1064nm.

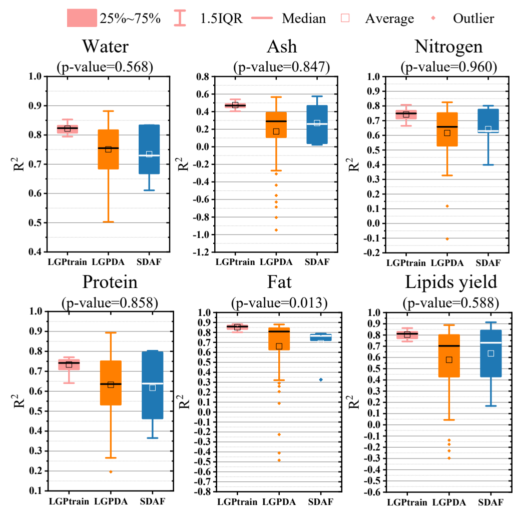

Figure 2: Training (light red) and test performance (orange for LGP and blue for SDAF) over 10 times six-fold cross-validation.

4.2 Example DAF Designed by LGP

We take an example DAF designed by LGP as a case study to analyze the newly designed DAFs. For the sake of simplicity, we select the DAF with the best training performance over the first folds in the cross-validation for water as the example DAF, as shown in Eq. (1).

$$\rho(\mathbf{x}_i) = \min(1, \mathbf{x}_i) + \texttt{rand} \quad (1)$$

In addition, we show the manually designed SDAF in Eq. (2) based on the terminals and functions of LGP for contrastive analysis.

$$\texttt{SDAF}(\mathbf{x}_i) = \mathbf{x}_i + \mathbf{x}_i \times \texttt{rand}_1 + \texttt{cursor} \times \texttt{rand}_2 + \texttt{rand}_3. \quad (2)$$

First, we can see that $\rho(\mathbf{x}_i)$ obtained the offset mainly based on $\mathbf{x}_i$, which is consistent with SDAF. Second, the newly designed DAF is more concise than SDAF. Although the example DAF has a high similarity with the manually designed SDAF, it at least implies that $\mathbf{x}_i$ and the random value rand are the essential components in SDAF for our fish spectroscopic data.

4.3 Regression Performance over Different Augmented Data

To verify the effectiveness and generality of the example DAF designed by LGP, we compare the regression performance on six different training sets, including five augmented training sets by different methods. Specifically, the six training sets include:

1) "original": the original training set,
2) SDAF: the augmented training set by SDAF [1],
3) "rand": the augmented training set by rand,
4) "mixup": the augmented training set by a mix-up method [12],
5) "chatGPT": the training set augmented by ChatGPT-4o [9],
6) "LGPDA": the augmented training set by our example LGP-designed DAF, as shown in Eq. (1).

"rand" is a crippled version of our method that directly adds a random offset ranging in $[-\alpha, \alpha]$ to $\mathbf{x}_i$. "mixup" is a common data

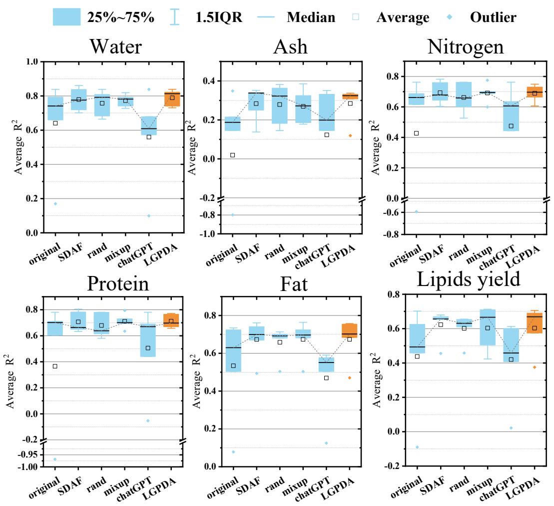

Figure 3: Average regression performance of machine learning models over different augmented data.

augmentation method in machine learning. It interpolates new data instances between randomly selected instances and their neighbors. “chatGPT” augments the training data by a generative Python program.

The chatGPT-generated Python program manipulates eight data augmentation techniques for generating new data, including adding and multiplying noises, randomly shifting, smoothening, and etc. The prompt for chatGPT and the full code of the data augmentation are shown in the supplement.

We perform six-fold cross-validations on the six training sets. The five augmented training sets consist of the original data and the newly generated data by the five DA methods, respectively. Each DA method produces approximately 500 new data instances, i.e., five times of the original training data.

We measure the effectiveness of a training set by the regression performance (test R^2) of five machine learning models. The five machine learning models include partial least square regression [11], K-nearest neighbor regression, multilayer perception regression [8], XGBoost regression [3], and support vector regression [10]. Each machine learning model runs 10 times of six-fold cross-validation. The average test R^2 of the machine learning models on the six training sets are shown in Fig. 3.

In Fig. 3, we can see that the regression performance over the four augmented training sets (i.e., SDAF, rand, mixup, and LGPDA) is better than the original training data and chatGPT. It confirms the effectiveness of the manually designed DAFs and LGPDA for fish spectral data. Particularly, LGPDA averagely helps the machine learning models achieve better performance than the other DA methods in many cases, such as water, ash, and protein. For the other targets such as fat and lipids yield, LGPDA is still very competitive with the other manually designed DAFs. Given that the example DAF (Eq. 1) is only trained on water, the results also imply a good generality of LGPDA to unseen biomass targets. The results show the promising performance of LGPDA in augmenting spectral datasets and enhancing machine learning performance.

5 Conclusions

This paper proposes to design new DA methods by GP. The results show that the proposed GP method can find more effective or at least competitive DAFs than the one designed by human experts in multiple runs. This indicates a high potential in enhancing machine learning models by automatically designed DA methods. The example DAF designed by the GP method allows us to understand and analyze its behaviors, which is important in real-world applications. The comparison among five popular DA methods confirms that the example DAF by our GP method has superior performance on most of the tested biomass targets.

This paper provides another perspective for generative artificial intelligence, that is, synthesizing generative models with random values to produce new data by GP methods. As GP has many unique advantages such as good interpretability and less dependence on a large number of training data, it will be interesting to promote GP as a specialized and customized generative model for certain domains.

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